

Pizza Sales Performance Analysis

Dashboard Documentation (2015 Data)

PROJECT DOCUMENTATION
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PIZZA SALES ANALYTICS |

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1. Project Objective and Business Problems

Objective

This documentation covers a Power BI dashboard (analysing_dashboard.pbix) built to analyse a full year of pizza-restaurant sales data (2015). It examines order-level transactions to surface revenue drivers, demand patterns, product performance, and ingredient economics, and pairs the dashboard's data model with the written findings captured in the accompanying analysis report.

Business Problems

The dashboard was built to answer:

- Which products and categories generate the most revenue and volume?
- How does demand fluctuate by hour, day of week, and month?
- How do pizza size and ingredient choices relate to sales performance?
- Which ingredients and menu items are the strongest and weakest contributors, and by how much?
- How does average pizza price relate to total volume across categories?
- How often do customers order more than one pizza, and how quickly do new orders arrive?

These answers support operational decisions such as staffing rosters, menu engineering, pricing tiers, and seasonal promotions.

2. Data Sources and Transformation

Data Source

The model is built from a single flat file, `data_pizza.csv`, loaded through Power Query with `Csv.Document`. Each row is one order line with the columns: `order_id`, `date`, `time`, `pizza_id`, `quantity`, `size`, `price`, `name`, `category`, and `ingredients` (a comma-separated ingredient list per pizza).

Key Transformations

Unlike a dashboard built directly on separate `orders` / `order-details` / `pizza` / `pizza-type` tables, this model derives every dimension from the one source file inside Power Query, then relates them back together in the data model:

- Promoted headers and set explicit data types (`order_id` as whole number, `date` as date, `time` as time, `price` as decimal, etc.).
- Added a custom "Size Order" column (`S = 1`, `M = 2`, `L = 3`, `XL = 4`, `XXL = 5`) so size labels sort logically instead of alphabetically.
- Built a full-year "Dim Date" calendar (1 Jan – 31 Dec 2015) from a generated list of dates, then added Month, three-letter Month Name, Start of Week, three-letter Day Name, Day of Week, Week of Year and Day of Year columns; "is Weekend" is a DAX calculated column (`SWITCH on WEEKDAY`).
- Built a "Dim_Time" table (Hour, Minute, Second, Time) to support hour-level analysis independently of the calendar.
- Derived the "Pizza" dimension (`pizza_id`, `name`, `category`) by removing the transaction columns from `data_pizza` and de-duplicating on `pizza_id`.
- Derived the "Pizza Ingredients" bridge table by splitting the ingredients text on commas (`Splitter.SplitTextByDelimiter`) and expanding it into one row per pizza/ingredient pair, then trimming whitespace.
- Derived the "Ingredients" dimension as the distinct list of individual ingredients across all pizzas.
- Built an "Order Date Time" table with two calculated columns — Next Order Time (`LOOKUPVALUE` against `order_id + 1`) and Minutes Differences (`DATEDIFF`, blank when the next order falls on a different day) — purpose-built to measure the gap between consecutive orders.
- Added disconnected parameter tables (Parameter, Parameter 2, "quantity/revenue", "quantity/revenue options", "top/bottom options") so single visuals can be toggled by the viewer between Quantity and Revenue, and between Top 5 and Bottom 5 rankings, without duplicating charts.

3. Data Model and DAX

Schema

- Fact table: data_pizza — one row per order line (order, date, time, pizza, quantity, size, price).
- Dimensions: Dim Date, Dim_Time, Pizza, Pizza Ingredients (bridge), Ingredients, Order Date Time.
- Disconnected control tables: Parameter, Parameter 2, quantity/revenue, quantity/revenue options, top/bottom options.

Relationships

From	To	Cardinality	Cross-filter
data_pizza[date]	Dim Date[Date]	Many-to-one	Single
data_pizza[pizza_id]	Pizza[pizza_id]	Many-to-one	Single
Pizza Ingredients[ingredients]	Ingredients[ingredients]	Many-to-one	Single
Pizza Ingredients[pizza_id]	Pizza[pizza_id]	Many-to-one	Both

The Pizza Ingredients → Pizza relationship is set to bidirectional filtering so that filtering by an ingredient (e.g. on the Ingredient Analysis page) correctly restricts the Pizza and data_pizza tables as well.

Key DAX Measures

A representative set of the measures created for this model:

Measure	Logic / purpose
Total Revenue	SUMX('data_pizza', data_pizza[price] * data_pizza[quantity])
Total Quantity Sold	SUM(data_pizza[quantity])
Total Orders	DISTINCTCOUNT(data_pizza[order_id])
Avg Order Value	DIVIDE([Total Revenue],[Total Orders])
Avg Pizza per Order	DIVIDE([Total Quantity Sold],[Total Orders])
Avg Pizza Price	DIVIDE([Total Revenue],[Total Quantity Sold])
Total Items per Order	CALCULATE([Total Quantity Sold], ALLEXCEPT(data_pizza, data_pizza[order_id]))
Quantity/Revenue	SWITCH on the "quantity/revenue options" parameter to let one visual show either measure
Pizza Rank	RANKX over all pizzas by Quantity/Revenue, direction driven by the Top 5 / Bottom 5 slicer

Measure	Logic / purpose
Ingredient Rank	RANKX of ingredients within the currently selected pizza category
Ingredient Quantity	SUMX('Pizza Ingredients', CALCULATE([Total Quantity Sold]))
Avg Ingredient Revenue / Quantity	AVERAGEX(ALL(Ingredients[ingredients]), ...) — the two reference-line averages used to build the ingredient quadrant chart
Previous Order Time / Avg Time Interval Between Orders	Looks up each order's timestamp against the prior order and averages the gap in minutes
Correlation measures (2)	Pearson correlation coefficient, computed manually in DAX, for (a) category price vs. quantity and (b) ingredient quantity vs. revenue

Notable calculated columns

Two calculated columns do meaningful classification work rather than simple lookups:

```
Order Quantity Status (on data_pizza)
VAR TotalQty =
    CALCULATE(
        DISTINCTCOUNT(data_pizza[pizza_id]),
        ALLEXCEPT(data_pizza, data_pizza[order_id])
    )
RETURN
SWITCH(TRUE(),
    TotalQty = 1, "Single-item",
    TotalQty > 1, "Multi-item",
    "Unknown")
```

This labels every order line as Single-item or Multi-item based on how many distinct pizzas appear in that order, and drives the order-mix pie chart.

```
Minutes Differences (on Order Date Time)
VAR CurrentDate = DATEVALUE('Order Date Time'[Date Time])
VAR NextDate = DATEVALUE('Order Date Time'[Next Order Time])
RETURN
IF(CurrentDate = NextDate,
    DATEDIFF('Order Date Time'[Date Time],
        'Order Date Time'[Next Order Time], MINUTE),
    BLANK())
```

This measures the gap to the next order only when both fall on the same day, avoiding an artificial overnight gap being counted as "time between orders".

4. Visuals and Key Insights

Visuals (by report page)

Page 1 — Common Metrics and Sales Trends

- KPI cards: Total Revenue, Total Quantity Sold, Total Orders, Avg Pizza per Order, Avg Order Value
- Line chart "Daily Sales Performance": Revenue, Orders and Quantity Sold across Day of Year / Week of Year
- Pivot-table heatmap "Sales Performance by Day of Week and Time of Day": Revenue by Hour × Day Name
- Month slicer and page-navigation buttons

Page 2 — Category and Pizza Analysis

- 100% stacked column chart: "Sales by Category and Size"
- Clustered bar charts by category and by pizza name, driven by field-parameter slicers so the viewer can switch between Quantity and Revenue, and between Top 5 and Bottom 5

Page 3 — Ingredient Analysis

- Category slicer plus most-popular / least-popular ingredient tables per category
- Scatter chart of Total Quantity Sold vs. Total Revenue by ingredient, with a correlation-coefficient card

Page 4 — Average Indicators

- Scatter chart of Avg Pizza Price vs. Total Orders by category, with its own correlation-coefficient card and trend line
- Pie chart of order mix (Single-item vs. Multi-item) and a card for the average time interval between orders

Key Insights Identified

KPIs

- Total Revenue: \$817.86K
- Total Quantity Sold: ~49.6K pizzas
- Total Orders: 21,350
- Average Order Value: \$38.31
- Average Pizzas per Order: 2.32 — most customers buy more than one pizza per order.

Sales trends and seasonality

Weekly revenue typically runs \$13K–\$17K, dipping toward the start and end of the year and peaking around week 48. By day of week (exact revenue, ranked):

Day	Total revenue
Friday	\$136.07K

Day	Total revenue
Thursday	\$123.53K
Saturday	\$123.18K
Wednesday	\$114.41K
Tuesday	\$114.13K
Monday	\$107.33K
Sunday	\$99.20K

Thursday–Saturday are the strongest days; Sunday and Monday are weakest. By hour, revenue peaks sharply at 12PM–1PM (lunch), with a secondary rise at 5PM–7PM (early dinner); very little revenue is generated before 11AM or after 9PM. By month, ranked from strongest to weakest:

Month	Total revenue
July (best)	\$72.56K
May	\$71.40K
March	\$70.40K
November	\$70.40K
...	...
February	\$65.16K
December	\$64.70K
September	\$64.18K
October (weakest)	\$64.03K

October, February, September and December are considered underperforming months. October is the single weakest month (\$64.03K), narrowly below September (\$64.18K), December (\$64.70K) and February (\$65.16K). The gap between all four is under 2%, so they behave as one prolonged low-demand cluster rather than four distinct dips.

Product performance

Category ranking by both volume and revenue, with average selling price added:

Category	Quantity sold	Revenue	Avg. price/pizza
Classic	14,888	\$220.05K	\$14.78
Supreme	11,987	\$208.20K	\$17.37
Chicken	11,050	\$195.92K	\$17.73
Veggie	11,649	\$193.69K	\$16.63

Classic leads on both volume and revenue but sells at the lowest average price (\$14.78); Chicken and Supreme carry the highest average prices (\$17.73 and \$17.37) despite lower volumes, confirming they are the model's premium categories rather than its highest-volume ones.

Top 5 pizzas by quantity sold:

Pizza	Quantity sold	Revenue
The Classic Deluxe Pizza	2,453	\$38.18K
The Barbecue Chicken Pizza	2,432	\$42.77K
The Hawaiian Pizza	2,422	\$32.27K
The Pepperoni Pizza	2,418	\$30.16K
The Thai Chicken Pizza	2,371	\$43.43K

Bottom 5 pizzas by quantity sold:

Pizza	Quantity sold	Revenue
The Brie Carre Pizza	490	\$11.59K
The Mediterranean Pizza	934	\$15.36K
The Calabrese Pizza	937	\$15.93K
The Spinach Supreme Pizza	950	\$15.28K
The Soppressata Pizza	961	\$16.43K

The Brie Carre Pizza is the clearest menu-engineering candidate: it sells roughly a fifth of the volume of the next-lowest pizza and generates under \$12K in revenue for the year, well below every other item on the menu.

Size and ingredient analysis

- Large is the single biggest size by volume, followed by Medium then Small; XL and XXL together make up a small fraction of sales and are effectively niche sizes.
- Garlic is the most popular ingredient across multiple categories (a genuine cross-category favourite), while Red Peppers, Red Onions and Chicken are each the top ingredient within their own category.
- Pesto Sauce, Green Peppers, Kalamata Olives, Plum Tomatoes and a small cluster of premium toppings (Caramelized Onions, Pears, Prosciutto, Thyme, Brie Carre Cheese) are consistently the least-used ingredients.

Correlations

- Ingredient quantity vs. revenue: correlation ≈ 1 — expected, since revenue is derived directly from quantity (Revenue = Price $\times$ Quantity), so this measures a mathematical dependency rather than an independent relationship.
- Average pizza price vs. total quantity sold, across the four categories: correlation ≈ -0.95 — a strong negative relationship. Classic (the cheapest category) sells the most; Chicken and Supreme (the priciest) sell the least, confirming clear price sensitivity in the customer base.

Order frequency and timing

Of the 21,350 orders placed in the year, 8,201 (38.4%) contained only a single distinct pizza and 13,149 (61.6%) contained more than one, which means multi-item orders are more in favour of customers.

Single-item orders account for only about 16.7% of all pizzas sold (~8.3K of ~49.6K), while multi-item orders account for about 83.3% (~41.3K). In other words, multi-item orders don't just outnumber single-item orders but also dominate sales volume even more heavily than the order-level split suggests, which strengthens the case for combo and bundle promotions.

The average interval between consecutive same-day orders is 11.0 minutes, indicating steady, continuous demand through service hours rather than isolated bursts.

Recommendations

- Pricing: introduce a mid-range tier between Classic and the premium Chicken/Supreme categories to capture volume without eroding the margin the -0.95 price–volume correlation implies is available.
- Promote top sellers (Classic Deluxe, Barbecue Chicken, Thai Chicken) and bundle them with drinks or sides at checkout.
- Reassess or reformulate the bottom performers — Brie Carre, Mediterranean and Calabrese — and reduce reliance on their associated low-volume ingredients (Prosciutto, Thyme, Pears, Caramelized Onions).
- Roster more staff for the confirmed peaks (12–1PM lunch, 5–7PM dinner, Thu–Sat) and consider trimming or discounting the quiet pre-11AM / post-9PM hours.
- Run targeted offers in the Oct–Sep–Dec–Feb low-demand cluster rather than treating each month separately, since the four behave as one extended trough.
- Lean further into combo and family-bundle offers: multi-item orders already drive roughly five out of every six pizzas sold, so incremental gains here have an outsized effect on total volume.

5. Interactive Features

Filters / Slicers

- Month slicer (Page 1) and pizza-category slicer (Pages 2 and 3)
- Field-parameter slicers for Quantity vs. Revenue and for Top 5 vs. Bottom 5 ranking (Page 2)
- Page-navigator buttons on every page for one-click movement between the four report pages

Interactivity

- Clicking any slicer cross-filters every visual on that page, including the KPI cards.
- The Quantity/Revenue and Top 5/Bottom 5 parameters let one pair of bar charts serve four different views without duplicating visuals.
- The Day-of-Week $\times$ Hour heatmap lets a viewer pinpoint the exact peak windows within a selected month.
- Both scatter charts carry a live correlation-coefficient card and trend line that recompute as filters change.
- The bidirectional Pizza Ingredients $\leftrightarrow$ Pizza relationship means selecting an ingredient filters back to which pizzas and categories use it, not just the reverse.